

BREATHE SMART! INVESTIGATING WEARABLE TECHNOLOGY FOR DETERMINING THE FIRST VENTILATORY THRESHOLD DURING INCREMENTAL EXERCISE

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INTRODUCTION

Background: The first ventilatory threshold (VT1) represents a key physiological transition during incremental exercise, where ventilation begins to increase disproportionately relative to oxygen consumption. Identifying VT1 can support individualised exercise intensity prescription and monitoring in endurance athletes. Traditionally, VT1 is determined using cardiopulmonary exercise testing (CPET), which requires specialised laboratory equipment and trained personnel [1]. This gold-standard method requires expensive gas-analysis equipment, technician expertise, and laboratory access, restricting its use to elite and well-funded programs.

Problem: Access to laboratory-based physiological testing can be limited for sub-elite athletes and smaller sporting programs due to equipment, expertise and financial requirements. This leads to relying on generic heart-rate zones rather than individualised thresholds, despite evidence that threshold-based training produces superior fitness outcomes [2]. Although recent research [3] highlights the potential of respiratory wearables for estimating ventilatory thresholds, commercially available devices remain largely unvalidated against gold-standard measures..

Solution: This study will investigate whether the Tymewear VitalPro chest strap [4] can identify VT1 during an incremental exercise test by analysing wearable- against ergospirometry-derived VT1 in sub-elite athletes. If acceptable agreement is demonstrated, this approach may provide a more accessible method for physiological assessment in sub-elite sport.

AIM

To determine whether respiratory data from Tymewear VitalPro breathing strap can identify VT1 during an incremental cycling exercise test in sub-elite athletes compared to compared with laboratory-based CPET.

HYPOTHESIS

It is hypothesised that VT1 determined using respiratory data from the Tymewear VitalPro will demonstrate acceptable agreement with VT1 determined using laboratory-based CPET.

STUDY DESIGN

A laboratory-based validation study will compare VT1 determined from the Tymewear VitalPro with laboratory-based CPET during an incremental cycling exercise test. Both systems will record respiratory and cardiovascular responses simultaneously.

PARTICIPANTS

Convenience sample of sub-elite endurance athletes (target $n = 20-30$) including cyclists, triathletes and runners, with no cardiorespiratory, metabolic, musculoskeletal, or neoplastic disease history.

METHODS

Participants will complete an incremental cycling exercise test to voluntary exhaustion while simultaneously wearing the Tymewear VitalPro and laboratory CPET equipment. Exercise workload will increase progressively until exhaustion, followed by a recovery period.

STATISTICAL ANALYSIS

Agreement between Tymewear-derived and laboratory-derived VT1 will be assessed using Bland-Altman analysis, including mean bias and 95% limits of agreement. Intraclass correlation coefficients and correlation analyses will also be used to assess the relationship between methods. Agreement will be examined using physiological outcomes including workload and heart rate at VT1.

EXPECTED OUTCOMES AND SIGNIFICANCES

If the Tymewear VitalPro demonstrates acceptable agreement with gold-standard ergospirometry, this study will provide the first independent validation of the device for VT1 detection, offering sub-elite athletes and community-level coaches a field-based alternative to laboratory testing for individualised training-zone prescription. If agreement is poor or systematically biased, the findings will still hold practical value clarifying the device's limitations and informing coaches on appropriate caution when using it to guide training intensity.

REFERENCES

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